

Assignment-5

1) Mountain car - v0

from the output:

* Number of steps = 200

* Rewards at each step = -1

Total reward

$$R_{total} = R_1 + R_2 + R_3 + \dots + R_{200}$$

since every step gives -1 reward:

$$R_{total} = (-1) + (-1) + \dots + (-1)$$

There are 200 steps:

$$R_{total} = 200 \times (-1)$$

$$R_{total} = -200$$

$$R_{average} = \frac{R_{total}}{\text{No. of steps}}$$

$$R_{average} = \frac{-200}{200}$$

$$R_{average} = -1$$

$$0 = \text{reward}$$

Mountain car Result

<u>Parameter</u>	<u>Value</u>
No. of steps	200
Total reward	-200
Average reward	-1

2) Frozen Lake - V1

Step Action Reward

1 3 0

2 2 0

3 2 0

4 0 0

5 0 0

6 1 0

7 0 0

Total Reward

$$R_{total} = 0 + 0 + 0 + 0 + 0 + 0 + 0$$

$$R_{total} = 0$$

Average Reward

Number of steps = 7

$$R_{average} = \frac{R_{total}}{\text{No. of steps}}$$

$$R_{average} = \frac{0}{7}$$

$$R_{average} = 0$$

Frozen Lake Result

Parameter

Value

Value

Value

No. of steps

000

7

29512 0.01

Total Reward

000

0

00000 0.01

Average Reward

1

0

00000 0.01

3) Cart Pole - v1

Reward = 1.

Total reward

$$R_{\text{total}} = 1 + 1 + \dots + 1$$

$$\boxed{R_{\text{total}} = N}$$

Average Reward

$$R_{\text{average}} = \frac{N}{N}$$

$$\boxed{R_{\text{average}} = 1}$$